

HEART VALVES

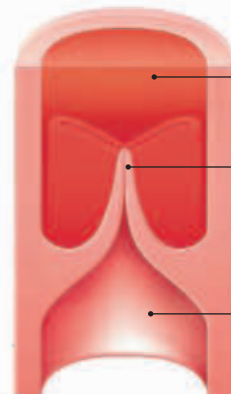
The heart is equipped with a set of valves to ensure that blood always flows in the right direction. On each side of the heart, a valve divides the upper atrium and the lower ventricle. This keeps the contracting ventricle from forcing blood back up into the atrium. Instead, it forces the blood through a semilunar (meaning "half-moon") valve (right), and into one of the arteries leaving the heart. The valve closes when the ventricle relaxes.



The blood surges forward.

The semilunar valve is forced open by the pressure of blood from behind.

The pressure of blood is high as the heart pumps.

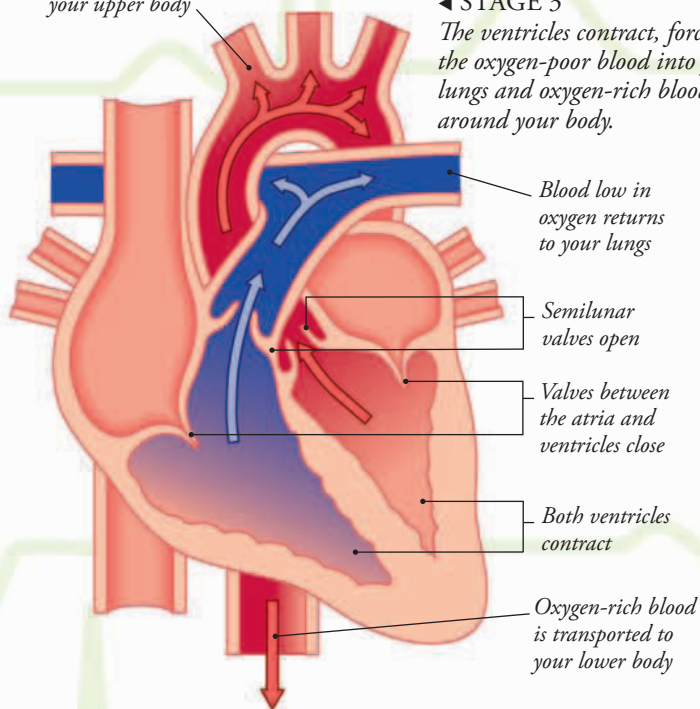


The blood cannot flow backward.

The semilunar valve is forced to close by the pressure of blood in front of the valve.

The pressure of blood drops as the heart rests between beats.

Blood rich in oxygen flows to your upper body



◀ STAGE 3

The ventricles contract, forcing the oxygen-poor blood into your lungs and oxygen-rich blood around your body.

Blood low in oxygen returns to your lungs

Semilunar valves open

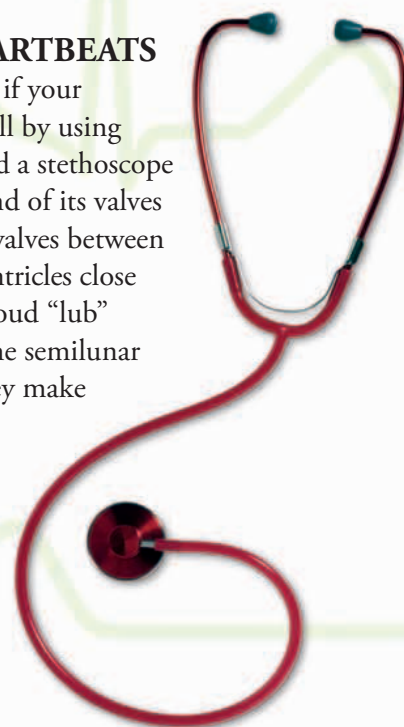
Valves between the atria and ventricles close

Both ventricles contract

Oxygen-rich blood is transported to your lower body

HEARING HEARTBEATS

A doctor can check if your heart is working well by using an instrument called a stethoscope to listen to the sound of its valves closing. When the valves between the atria and the ventricles close they make a long, loud "lub" sound, and when the semilunar valves slam shut they make a shorter, sharper "dup" sound.



FEELING A PULSE

As your heart pumps blood through your arteries, they expand and then spring back in time with every heartbeat. We call this a pulse. It is most obvious where an artery passes over a bone near the surface, such as in your wrist or neck. You can feel your pulse by pressing down on the artery on the inside of your wrist.

